

For Your Physician

The Leiomyosarcoma Project (LMSproject) is part of Count Me In, a nonprofit organization that brings together patients and researchers as partners to accelerate discoveries in cancer research. Count Me In is stewarded by the Broad Institute of MIT and Harvard, a leading nonprofit biomedical research institution; the Dana-Farber Cancer Institute, a leading cancer hospital; and Emerson Collective, a California-based social change organization. Count Me In's Leiomyosarcoma (LMS) Project is supported by a grant from the National Cancer Institute's Cancer Moonshot program and is part of the Participant Engagement and Cancer Genome Sequencing (PE-CGS) Network of research centers. By partnering directly with patients, we can accelerate discoveries in leiomyosarcoma, leading to the development of new therapies and treatment strategies.

Do participants receive individual results back from this study? If a participant elects to share archive tissue samples with the project, we may be able to share information about what we learned from the genomic sequencing of the tumor sample back with the participant. This information is intended to highlight how participant sample(s) are contributing to research and will not have implications for their treatment or prognosis. Because of regulatory restrictions, we are unable to share results with participants in New York and Canada.

The project is also partnering with Genome Medical and Invitae in order to sequence and share information on germline DNA (or “normal” DNA) from saliva samples. During the project, participants from whom we have received a blood or saliva sample and for whom we have received a tumor sample from a treating institution will be able to decide if they would like to provide an additional saliva sample to learn more about their normal DNA and speak with a genetic counselor to understand these results. If a participant does not participate in this process, their data will not be shared with Genome Medical and Invitae. This process is only available to eligible U.S.-based participants (within the 50 states).services there.

What makes this project different from existing studies?

The Leiomyosarcoma Project seeks to empower patients and families to accelerate cancer research by sharing their samples, clinical information, and experiences with the research community. Most patients are cared for in community settings where genomic studies are not typically conducted, so this study connects leiomyosarcoma patients around the country with genomics research performed at the Broad Institute, allowing them to participate regardless of where they live. The study was designed in partnership with leiomyosarcoma patients, families, advocates, and advocacy groups, who continue to provide input on future directions.

How is the genomic and clinical data shared with the research community?

Next generation sequencing (whole exome and transcriptome sequencing) is performed on tumor and germline DNA. Sequencing data are linked to de-identified clinical information, and the resulting data are used to identify drivers of tumorigenesis, mechanisms of response and resistance to therapies, and diagnostic, prognostic, and therapeutic biomarkers. The database of clinically annotated genomic information is shared on publicly available platforms such as cBioPortal and the NIH.

If you have any questions about this study, please reach out to us at info@lmsproject.org or **651-403-5556**.

What kinds of questions is this project aiming to address?

Over the past decade, genomic characterization of tumors has shed enormous light on the molecular underpinnings of cancer. These discoveries have led to the development of novel therapies and preventive measures that have already revolutionized cancer care. Despite this progress, the genomics of leiomyosarcoma remains poorly understood.

The types of questions we strive to answer include:

- What are all the genes and mutations that can lead to leiomyosarcoma?
- What explains why some patients show extraordinary responses to a particular treatment?
- What explains why some tumors never respond to a particular treatment?
- How can we develop better treatments for leiomyosarcoma?

Despite the progress that has been made to begin to answer these questions, we remain far from the goal. To get there, the detailed genomic characterization of many thousands of clinically annotated cancer samples will be required.

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